

Project Initialization and Planning Phase

Date	10 JULY 2026
Team ID	
Project Title	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution)

The **Emotion Detection & Learning Support Engine** is an AI-powered web application developed to detect users' emotions from text and provide personalized learning support. The system combines a

BiLSTM deep learning model and a **fine-tuned BERT transformer model** to classify emotions accurately. Based on the detected emotion, the application uses the **Google Gemini API** to generate personalized motivational messages, study tips, and learning recommendations.

The proposed solution integrates natural language processing, deep learning, and generative AI to improve students' emotional well-being and learning experience. The application provides real-time emotion analysis through a user-friendly Streamlit interface and supports educational decision-making through personalized AI assistance.

Key features include emotion detection, AI-generated learning recommendations, keyword enhancement, emotion history tracking, visualization of prediction results, and support for multiple emotion classes.

Project Overview	
Objective	The primary objective of this project is to develop an intelligent emotion detection system capable of accurately identifying human emotions from textual input using hybrid deep learning models. The system aims to support students and learners by providing personalized learning guidance, motivational suggestions, and emotion-aware educational assistance through artificial intelligence.
Scope	The project focuses on collecting and preprocessing emotion datasets, training BiLSTM and BERT-based deep learning models, classifying user emotions into multiple categories, generating AI-powered learning recommendations using Google Gemini API, and presenting the results through an interactive Streamlit web application. The solution supports educational institutions, students, researchers, and individuals seeking emotion-aware learning assistance
Problem Statement	
Description	Understanding a student's emotional state during learning is difficult using traditional educational systems. Existing platforms provide generic learning recommendations without considering the user's emotions. Manual identification of emotional conditions is time-consuming and inaccurate, reducing learning effectiveness and personalized support.
Impact	Failure to identify emotions may result in poor learning outcomes, reduced motivation, increased stress, and lack of personalized educational support. An AI-powered emotion detection system helps improve student engagement, emotional well-being, learning efficiency, and overall educational performance through intelligent recommendations.
Proposed Solution	

Approach	Develop an AI-based Emotion Detection & Learning Support Engine that combines BiLSTM and fine-tuned BERT models to classify emotions from user text. The application generates personalized educational guidance using the Google Gemini API and displays results through a Streamlit-based web interface with emotion visualization and learning recommendations..
Key Features	• Text-based emotion detection • Hybrid BiLSTM and BERT emotion classification • Detection of seven emotion categories (Joy, Sadness, Anger, Fear, Love, Surprise, Neutral) • Personalized learning recommendations using Google Gemini • Keyword enhancement for better emotion analysis • Emotion prediction history

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor	Intel Core i5 or higher
Memory	RAM specifications	8 GB
Storage	Disk space	20 GB Free SSD/HDD
Software		
Operating System	Platform	Windows 10/11 or Ubuntu

Programming Language	Development	Python 3.10+
Framework	Web Framework	Flask
Libraries	Machine Learning	Pandas, NumPy, Scikitlearn, Matplotlib, Joblib
IDE	Development Environment	VS Code
Development Environment	IDE	VS Code, Jupyter Notebook, Kaggle Notebook
Data		
Data	Hugging Face Dataset + Custom Neutral Samples	Emotion Dataset (Hugging Face – dair-ai/emotion) with additional Neutral class generated during training